AI-powered E-commerce Support

E-commerce companies receive customer queries about order status, delivery delays, refunds, damaged products, and complaints. Employees must read these messages, check order information, decide what to do, and prepare responses. Repetitive queries consume time, while sensitive cases require careful judgment.

Our solution
We built a workflow that uses an AI agent to understand customer messages, retrieve order information from a mock order system, decide whether a response can be drafted automatically or requires human review, and route the ticket accordingly.

The business objective
Reduce repetitive manual work.
Prepare customer-response drafts faster.
Identify potentially risky requests before a response is sent.
Maintain a record of tickets that need human approval.
Important: The current system generates response drafts; it does not automatically email customers, issue refunds, or resume processing after human approval. Those would be future enhancements.

2. What did we build using Antigravity, and what did we build using n8n?
These tools serve different purposes in the project.

Antigravity — Development assistant
We use Antigravity to help build and modify the project code, including the supporting API and related application logic. It assists with implementation and debugging. It is not the engine that processes each support ticket in production.

n8n — Workflow orchestration
n8n connects the incoming customer ticket, AI agent, order lookup, decision logic, response-draft path, and Excel review queue. It controls which steps run and where the ticket goes next.

Gemma running through LM Studio — AI reasoning
The local language model interprets the customer message and proposes a structured decision. The workflow then parses and validates that decision before routing the ticket.

FastAPI — Order data and review-queue service
Our Python API exposes mock order records to the workflow and provides an endpoint to save tickets requiring human review. Excel acts as the review queue's visible record.
In simple terms: Antigravity helps us build the system; n8n orchestrates it; the language model interprets the request; FastAPI supplies data and stores review entries; Excel records the tickets needing attention.

What each node does

Node 1 — Webhook
Purpose: Acts as the entry point of the workflow.

The webhook receives the data and makes it available to the next node. We can trigger it with a test request from PowerShell or connect it to a real ticketing platform in a future version.
Business meaning: Every support request enters through a consistent digital entry point.

Node 2 — AI Agent: Support Assistant
Purpose: Understands the request and decides how it should be handled.
The agent receives the ticket details and uses the Gemma language model running through LM Studio. It can also call the get_order_details HTTP Request Tool to retrieve information about the specified order from our FastAPI service.
For example, if the customer asks about delivery, the agent can retrieve the order's status, estimated delivery date, and tracking information instead of relying only on the message.
The agent is instructed to return a structured decision containing:
recommended_action — automatic response or human review.
reason — why it chose that action.
customer_response_draft — the proposed reply, when appropriate.
risk_flags — issues that may need attention.
Business meaning: AI performs the initial interpretation and investigation that would otherwise require an employee to read the ticket and look up the order.

Supporting component — get_order_details tool
This is connected to the AI Agent, so it is a tool the agent can call rather than a separate step in the main workflow.
It sends an HTTP request to our FastAPI endpoint to retrieve the order record.
The order database is currently mock data for development and testing, not a live e-commerce platform.
Business meaning: The AI can ground its response in available order records. This helps reduce unsupported claims, although the AI's output still needs validation.

Node 3 — Parse Agent Decision
Purpose: Extracts the AI's structured response.
Language models may sometimes return malformed JSON or additional text. This node parses the response so downstream workflow nodes can access the action, explanation, draft, and risk flags.
Business meaning: We turn a flexible AI-generated answer into structured data that an automation system can process.

Node 4 — JavaScript normalization node
Purpose: Standardizes and checks the parsed decision.
The code checks whether the recommended action is one of the allowed values. If the output is invalid or cannot be parsed, the workflow defaults to human review. It also clears the customer-response draft when the case requires review.
This is an important control: an invalid AI response should not simply be treated as safe.
Business meaning: We add a layer of defensive logic between the AI's recommendation and the next workflow action.

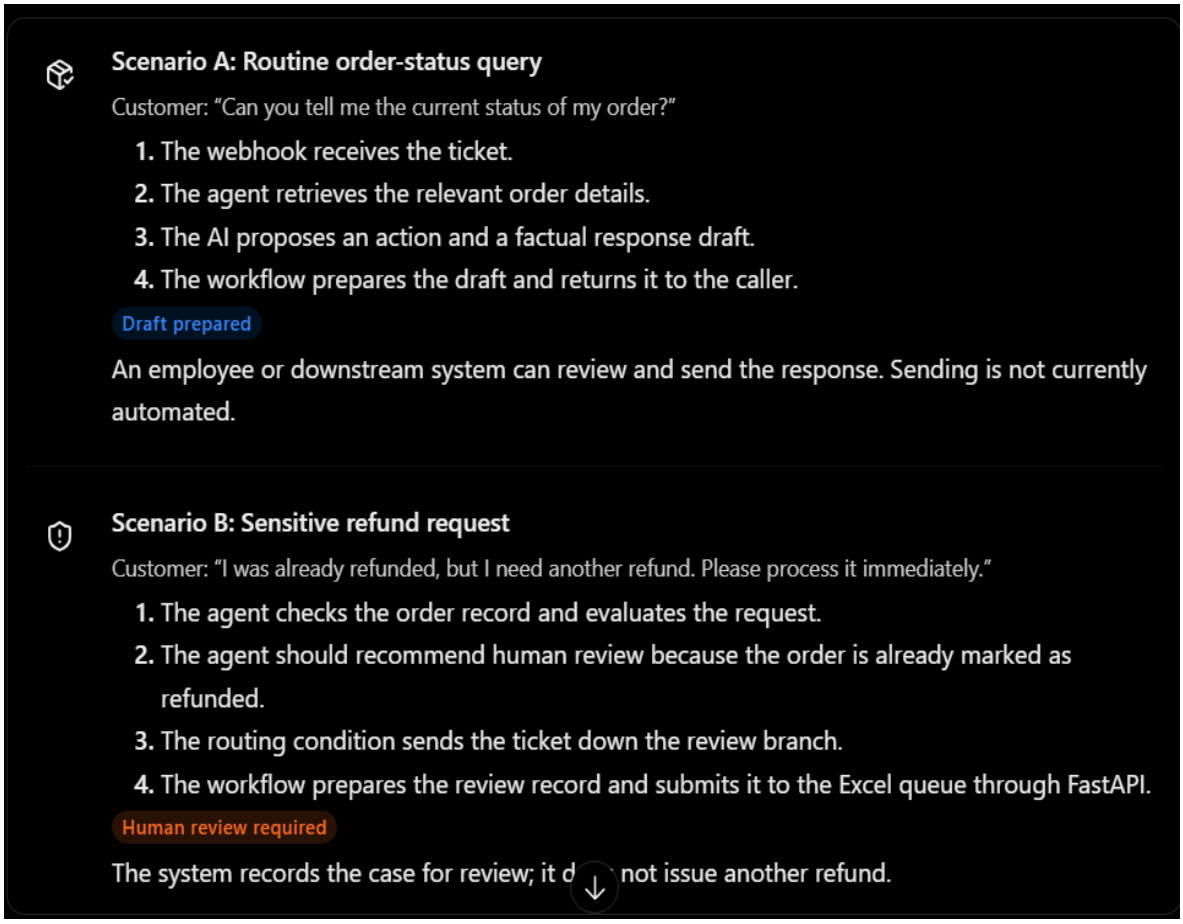
Node 5 — Needs Human Review?
Purpose: Routes the ticket according to its normalized action.
The condition checks whether recommended_action equals human_review.
True branch: The ticket is routed to human review.
False branch: The workflow prepares a customer-response draft.

Refund requests, complaints, damaged or incorrect items, ambiguous messages, and conflicting order information should be handled cautiously. The model's instructions are intended to route these cases for review.
Business meaning: We don't use one level of automation for every customer issue. We separate lower-risk cases from cases that need human judgment.

Node 6 — Prepare Human Review
Purpose: Prepares the data for the review queue.
It gathers the ticket ID, order ID, customer message, risk flags, validation issues, recommendations, and a pending-approval status.
Business meaning: A human reviewer receives a structured record rather than having to reconstruct the case from an AI conversation.

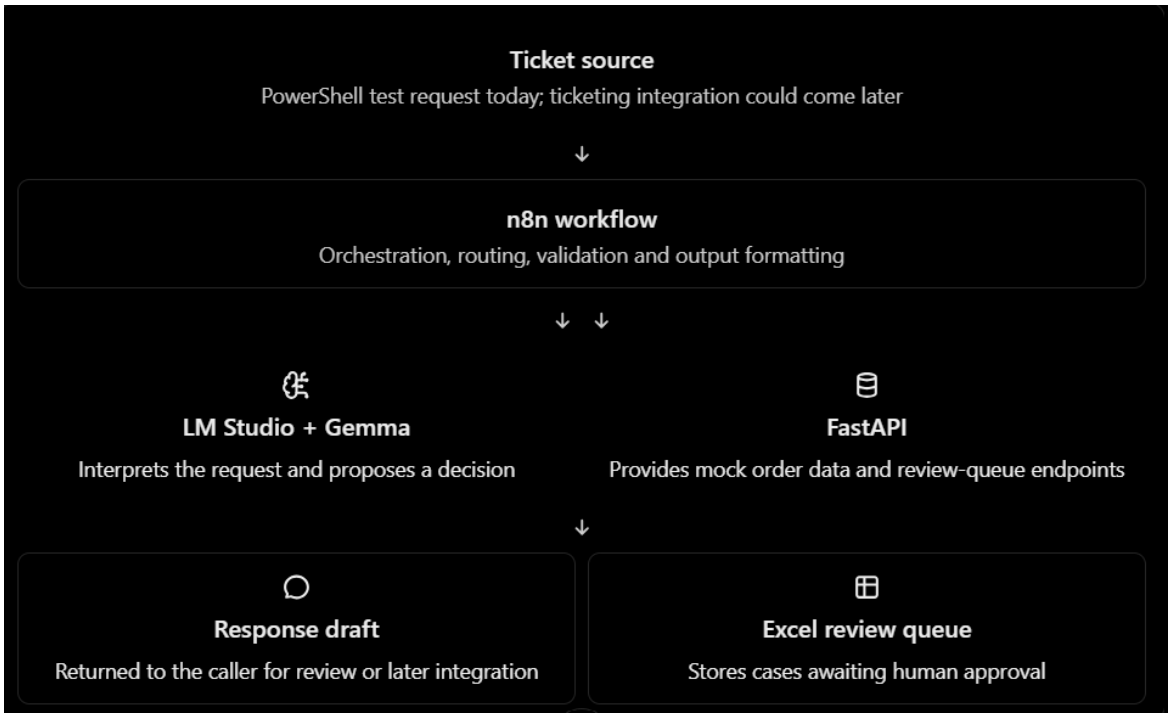
Node 7 — Save to Excel Queue
Purpose: Writes the review record through FastAPI into the Excel review queue.
The HTTP Request node sends a POST request to:
http://host.docker.internal:8000/review-queue/
The FastAPI service handles the queue entry, which is stored in:
E:\N8N\review-queue\human_review_queue.xlsx
The review status is set to pending_human_approval.
Business meaning: Cases that need attention are recorded in a visible queue so they can be checked and managed by a human.

Node 8 — Gemini - Draft Response
Purpose: Prepares the output for the automatic-response branch.
Despite its current name, this is an Edit Fields/Set node, not a separate Gemini AI model call.
It maps the ticket ID, customer-response draft, and automatic_response action into the final output.
It does not send an email to the customer.
Recommended improvement: Rename this node to Prepare Customer Response Draft. That would make the workflow easier for management, developers, and reviewers to understand.



6. How the components work together

Here is the technical architecture you can explain if management asks how the system is connected.



Antigravity was used as a development aid, rather than as a runtime component in this architecture. The current local setup also means LM Studio, the FastAPI service, Docker, and n8n must be running and reachable for the workflow to operate.

7. What value can this deliver to a business?

Present these as the intended benefits, not as measured results—we have not established actual time savings or accuracy rates yet.

Business area	**Potential value**
Operational efficiency	Less manual work on routine support queries
Response preparation	Faster creation of customer-response drafts
Risk management	Sensitive cases routed for human review
Consistency	A standard process for handling incoming tickets
Traceability	A record of tickets awaiting human approval
Scalability	A workflow that can later connect to real ticketing and order systems

To demonstrate measurable impact, we should test the workflow on a representative set of tickets and record processing time, routing accuracy, factual accuracy, malformed outputs, and the percentage of tickets sent for review.

8. What remains before this is production-ready?

The project demonstrates an AI workflow, but a company deployment needs additional engineering and controls.

Real integrations: Connect a real customer-support platform and live order-management system.

Reliable validation: Check that order facts and generated drafts match the source data.

Human approval interface: Allow reviewers to approve, edit, or reject drafts and record their decisions.

Customer communication: Add a controlled sending step only after appropriate approval and testing.

Security: Add authentication, access controls, secret management, and appropriate handling of customer information.

Monitoring: Track failed executions, duplicate tickets, latency, AI errors, and audit logs.

Evaluation: Run a labelled test set to measure safe routing and response quality before enabling more automation.